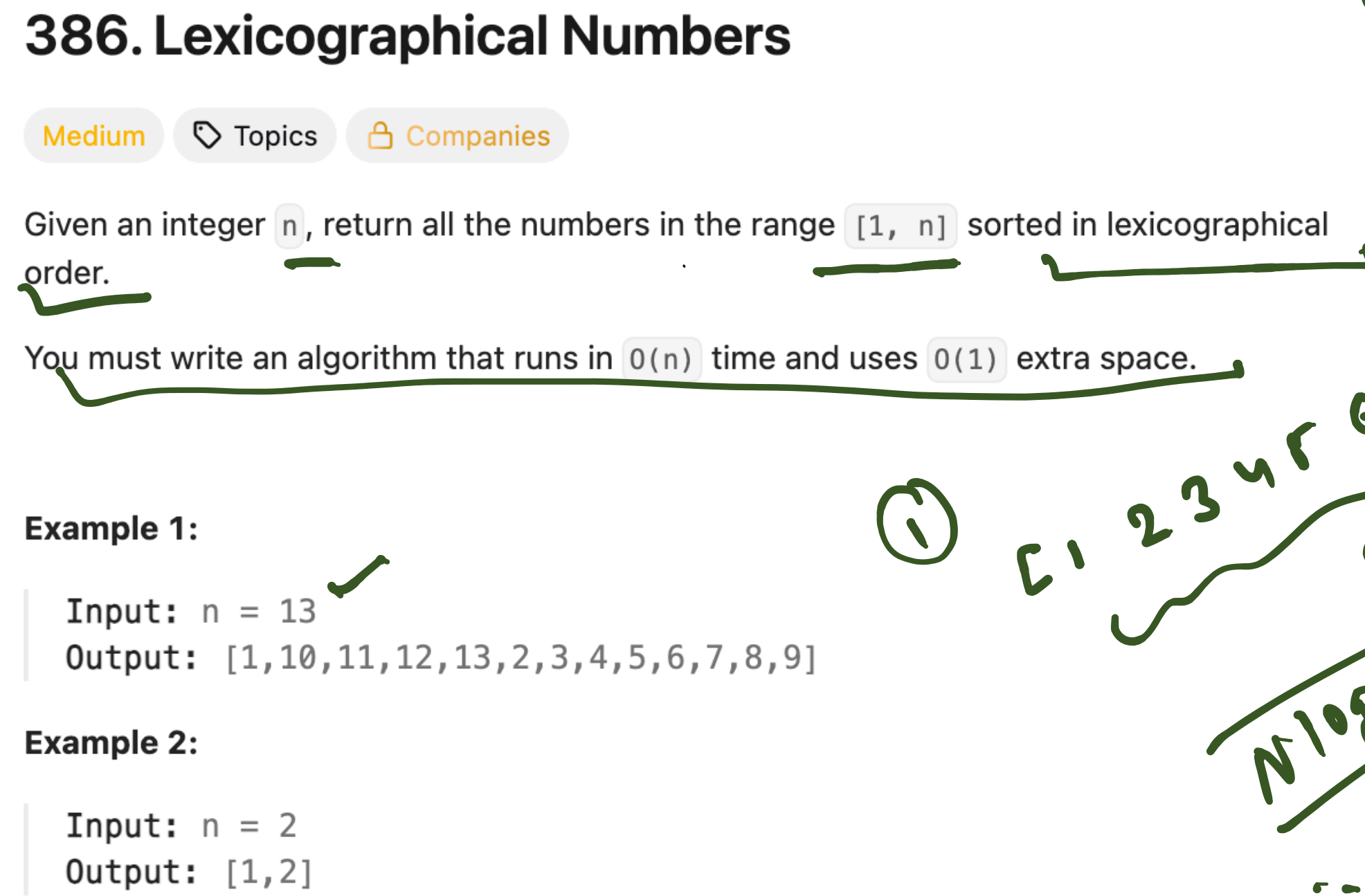
Lec-22

24/06/26 7:36 PM



386. Lexicographical Numbers

Medium · Topics · Companies

Given an integer `n`, return all the numbers in the range `[1, n]` sorted in lexicographical order.

You must write an algorithm that runs in `O(n)` time and uses `O(1)` extra space.

Example 1:

```
Input: n = 13
Output: [1,10,11,12,13,2,3,4,5,6,7,8,9]
```

Example 2:

```
Input: n = 2
Output: [1,2]
```

kunal > komal

① [1 2 3 4 5 6 7 8 9 10 11 12 13]

Sort

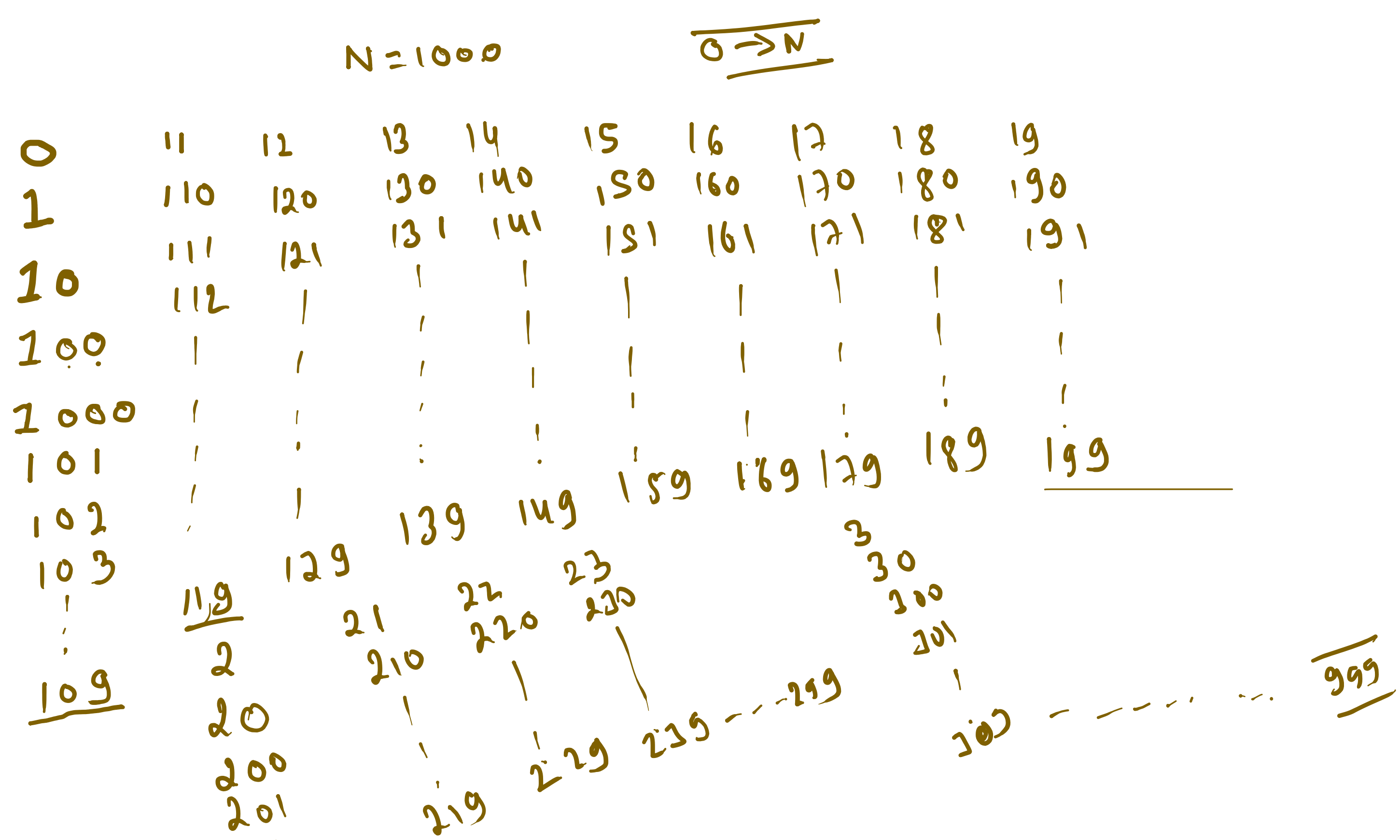
N log(N)

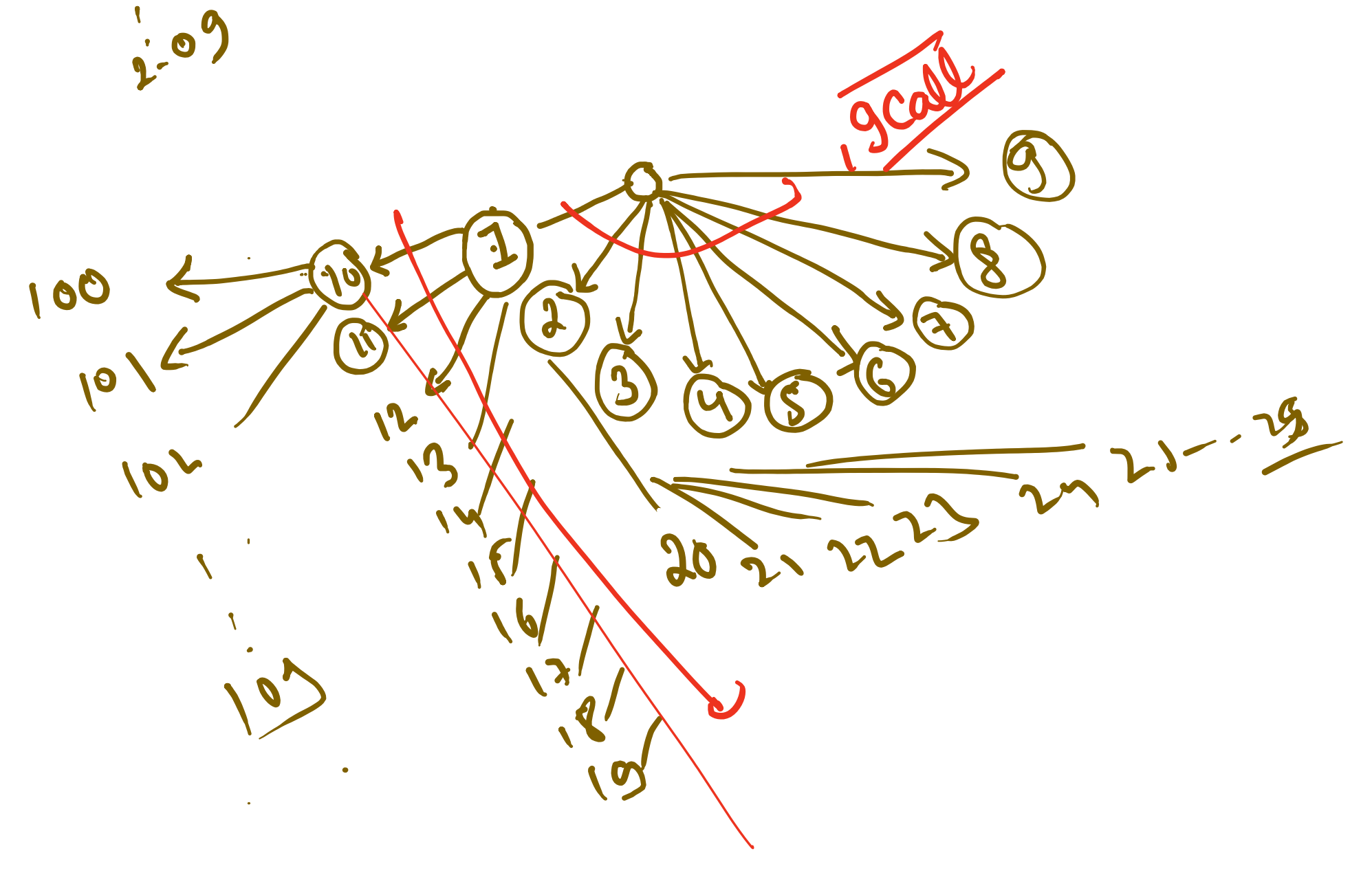
for(i= 1<=n i++){
ll.add(i)
}

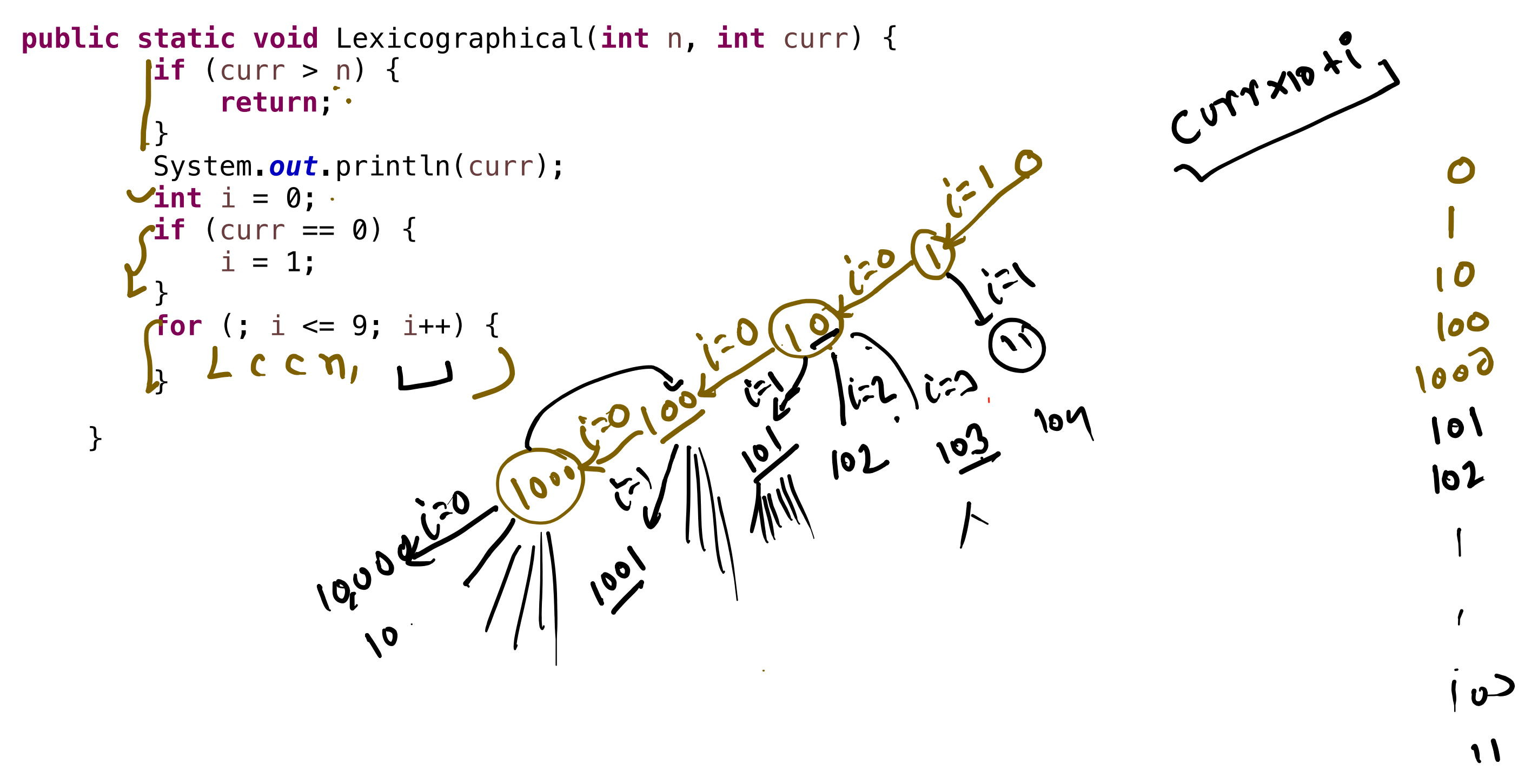
120 < 30

120 > 32

12 < 125

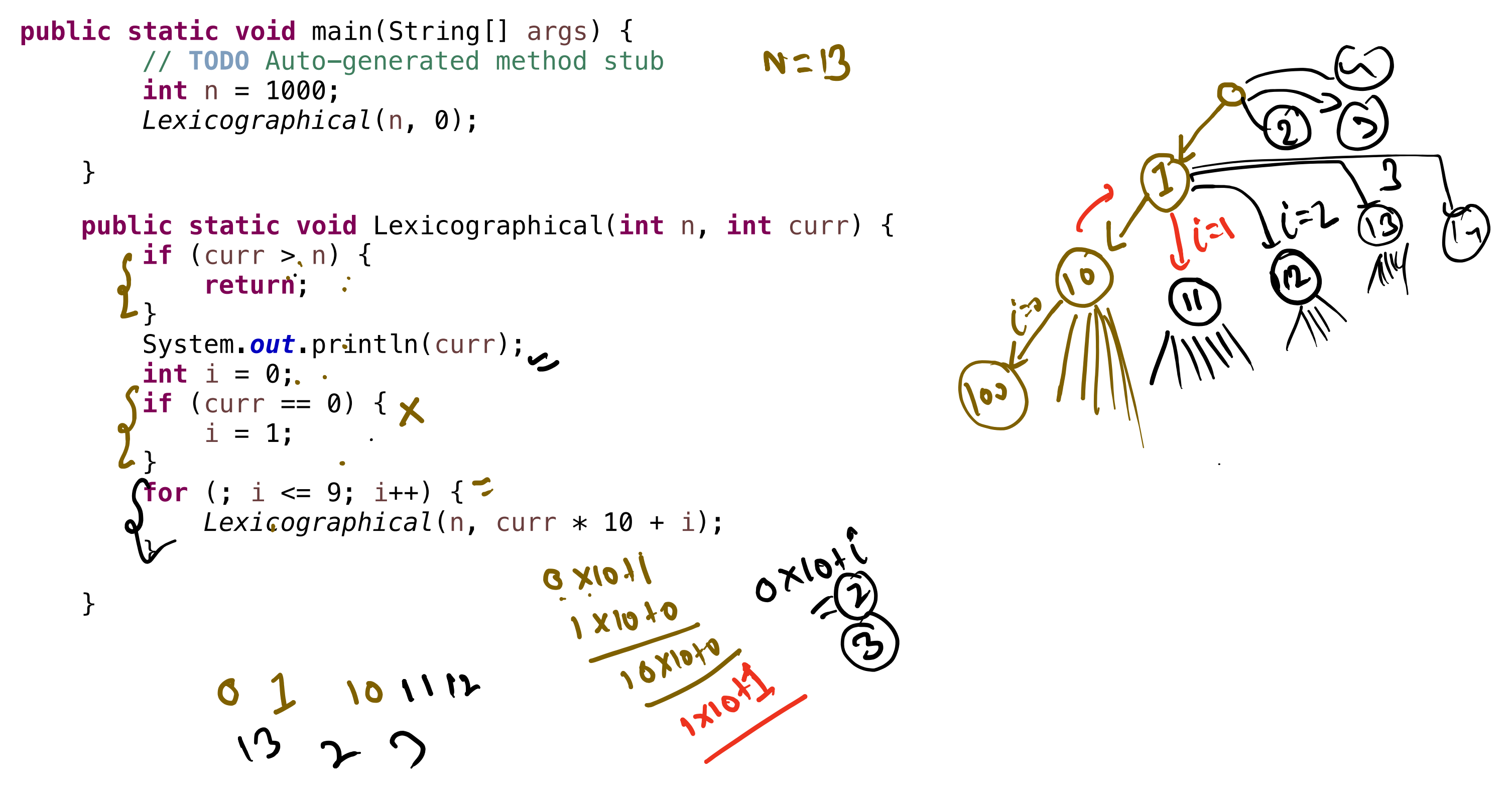

N=1000
0→N
0
1
10
100
1000
101
102
103
109
11 110 111 112 119
12 120 121 129
13 130 131 139
14 140 141 149
15 150 151 159
16 160 161 169
17 170 171 179
18 180 181 189
19 190 191 199
2 20 200 201 209
21 210 219
22 220 229
23 230 239 --- 299
3 30 300 301 309
999


9call
1
2
3
4
5
6
7
8
9
10
11
100
101
102
109
12
13
14
15
16
17
18
19
20 21 22 23 24 21 -- 29



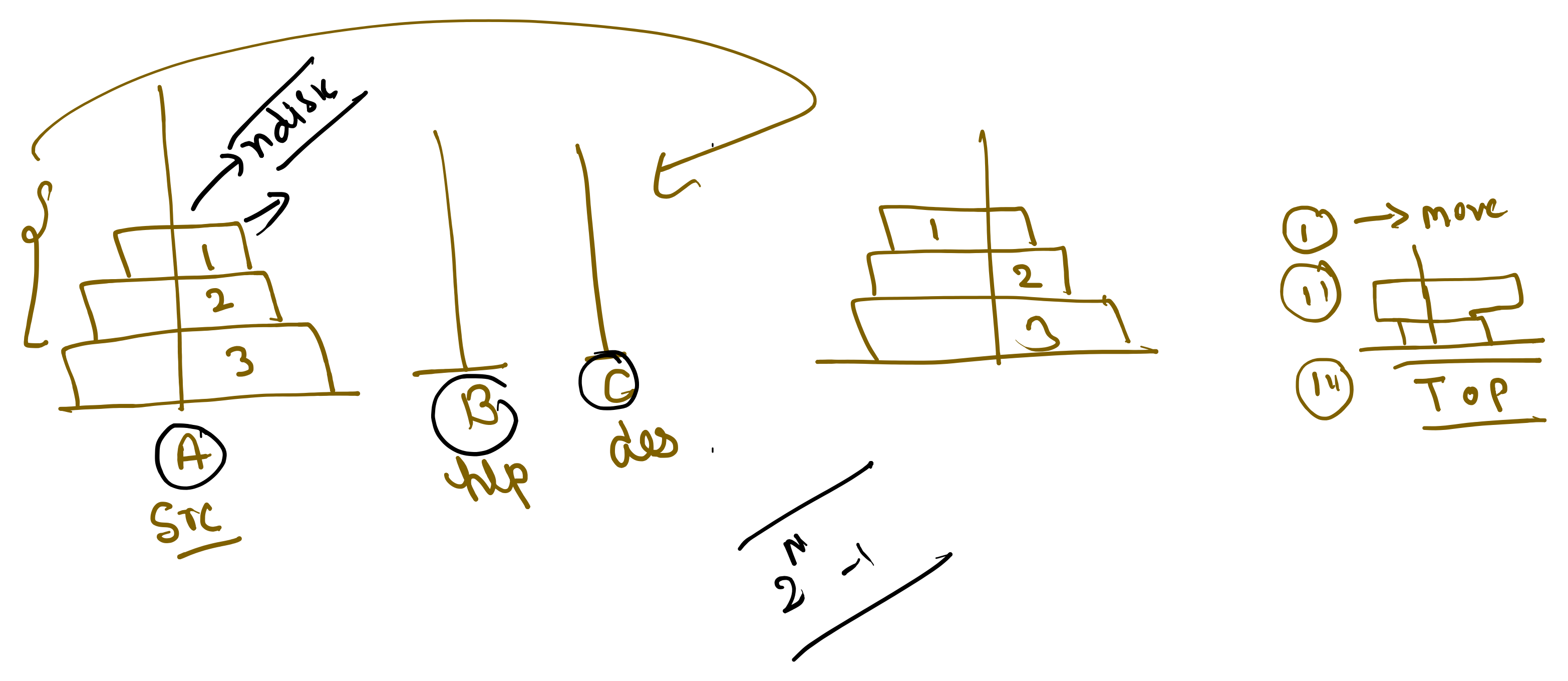
```
public static void Lexicographical(int n, int curr) {
	if (curr > n) {
		return;
	}
	System.out.println(curr);
	int i = 0;
	if (curr == 0) {
		i = 1;
	}
	for (; i <= 9; i++) {
	}

}
```

Lccn, ⌴)
curr×10+i
i=0
i=1
i=2
i=3
1
10
11
100
101
102
103
104
1000
10000
1001
0
1
10
100
1000
101
102
103
11

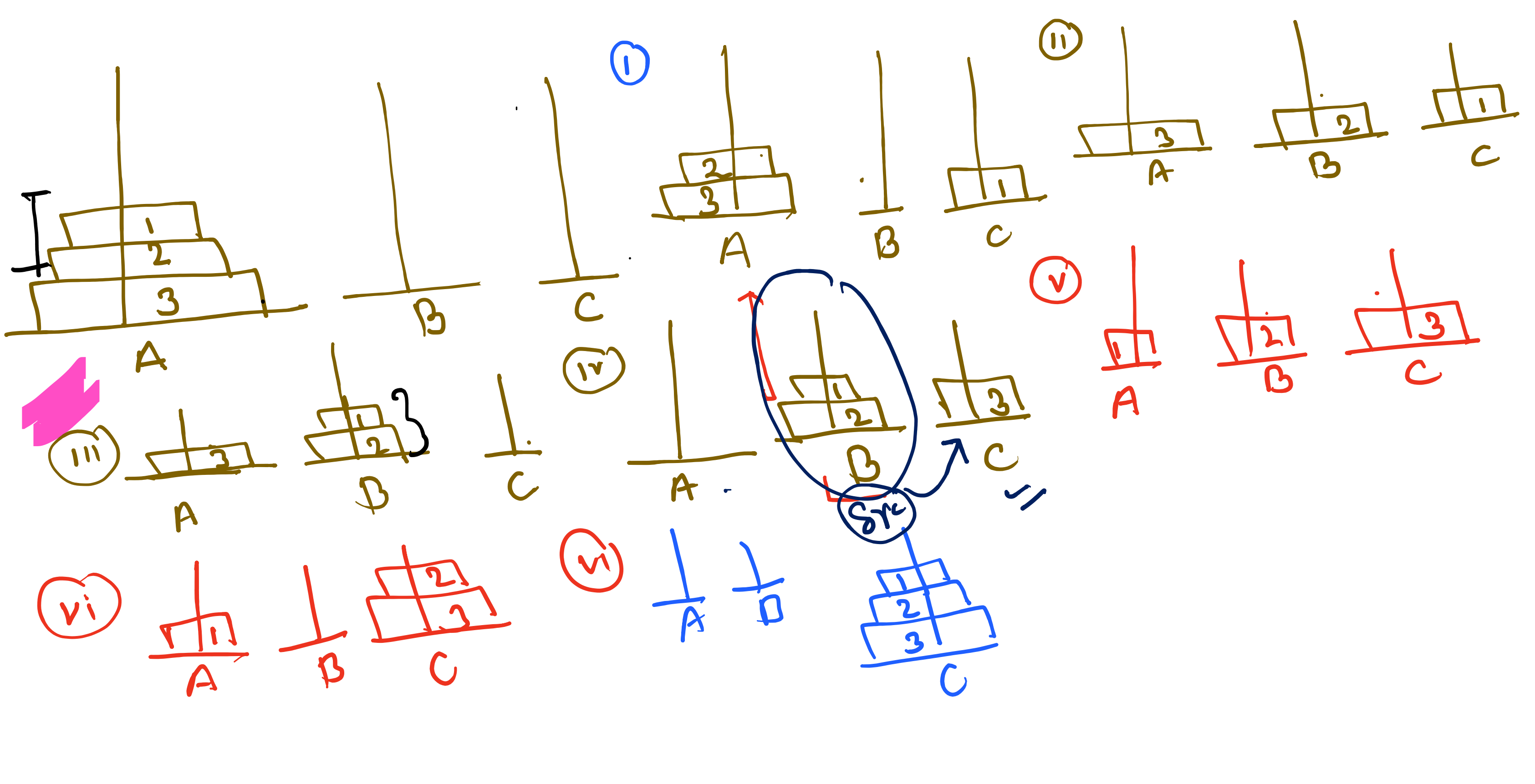


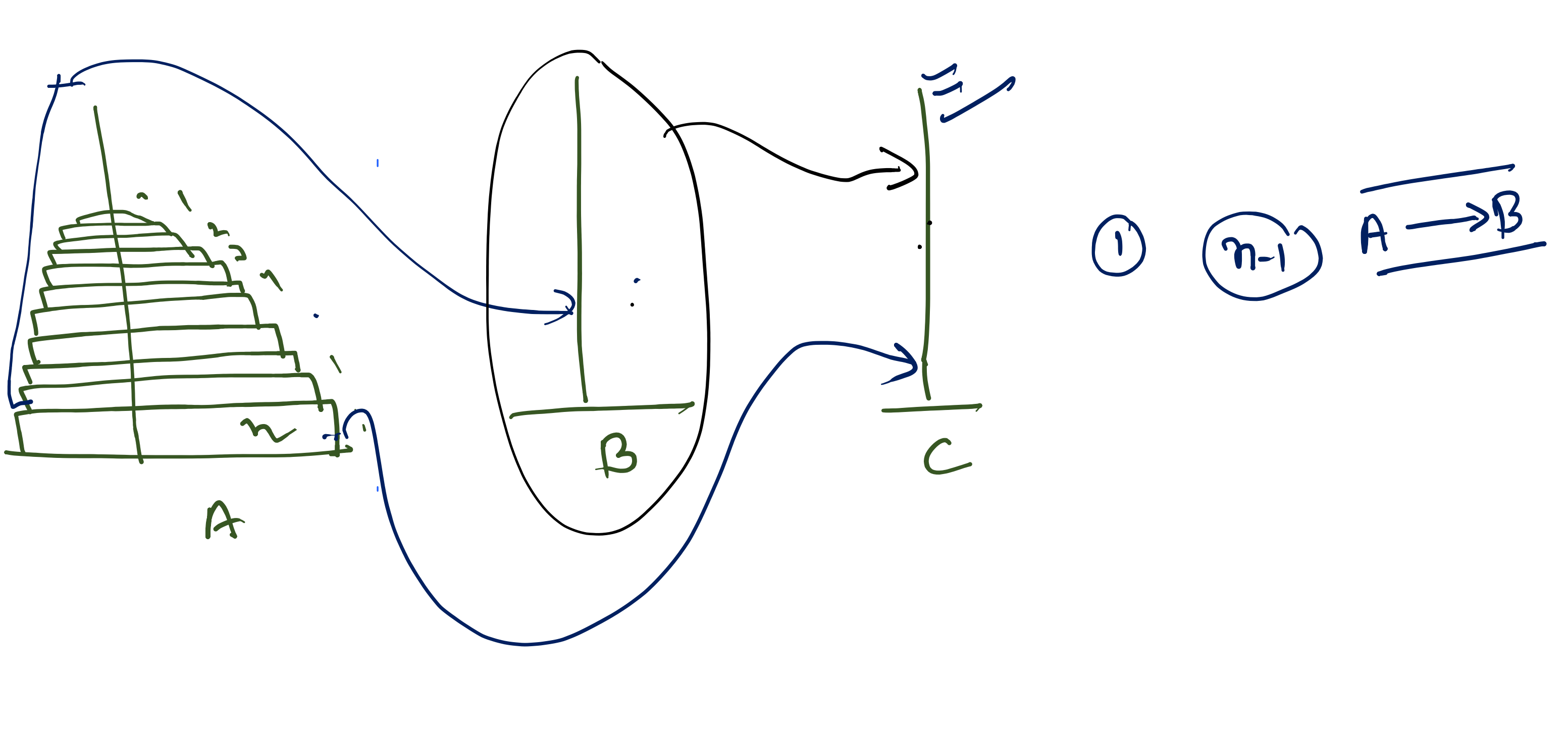
```
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 1000;
	Lexicographical(n, 0);

}

public static void Lexicographical(int n, int curr) {
	if (curr > n) {
		return;
	}
	System.out.println(curr);
	int i = 0;
	if (curr == 0) {
		i = 1;
	}
	for (; i <= 9; i++) {
		Lexicographical(n, curr * 10 + i);
	}

}
```

N=13
0 1 10 11 12
13 2 3
0×10+1
1×10+0
10×10+0
1×10+1
0×10+i
2
3
1
2
3
4
10
11
12
13
14
100
i=1
i=2

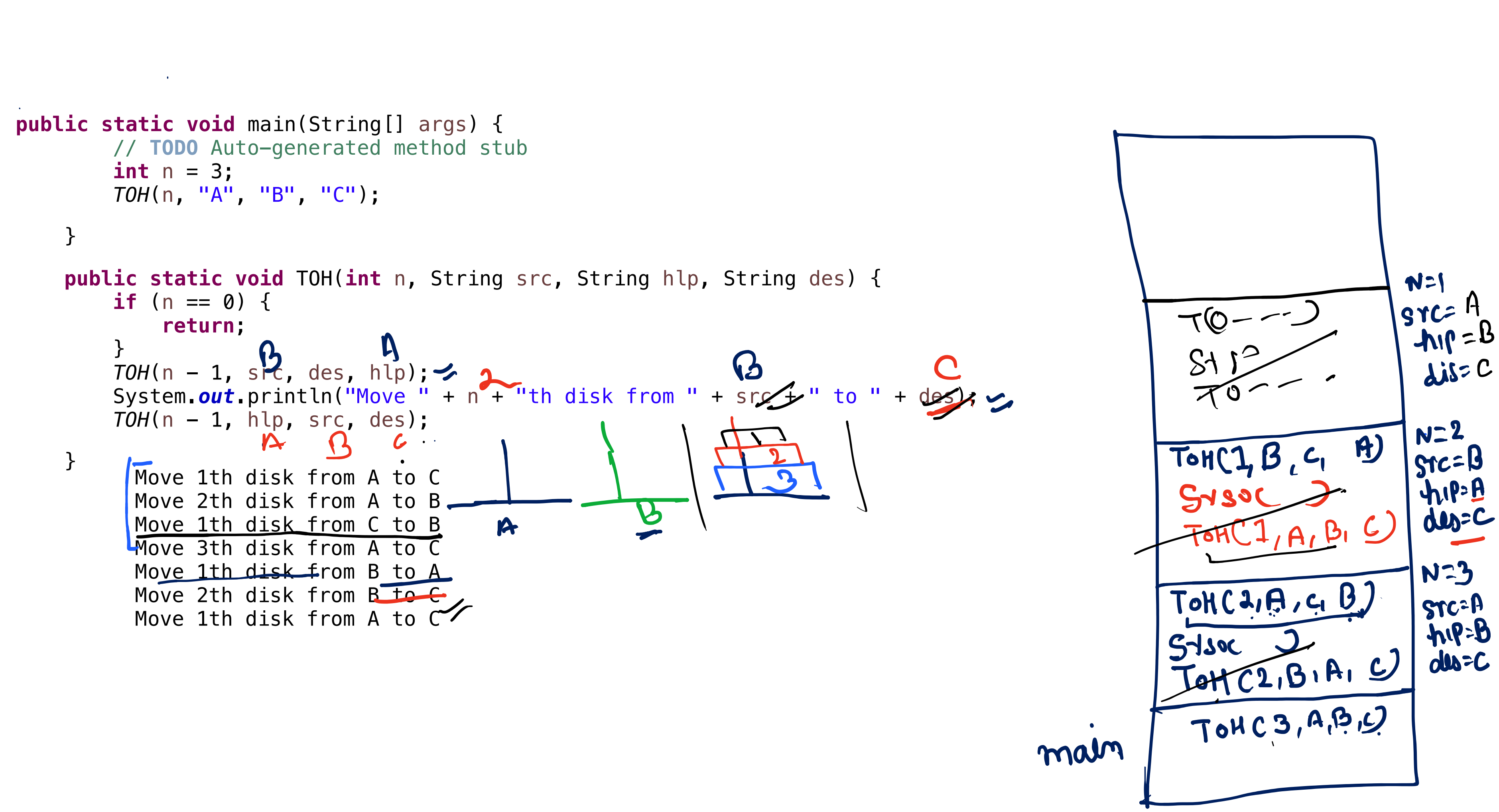

n disk
1
2
3
A
src
B
hlp
C
des
1
2
3
① → move
②
③
Top

$2^N - 1$


①
1
2
3
A
B
C
②
③
④
⑤
⑥
src
A
B
C



① (n-1) A → B



```
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 3;
	TOH(n, "A", "B", "C");

}

public static void TOH(int n, String src, String hlp, String des) {
	if (n == 0) {
		return;
	}
	TOH(n - 1, src, des, hlp);
	System.out.println("Move " + n + "th disk from " + src + " to " + des);
	TOH(n - 1, hlp, src, des);

}
```

```
Move 1th disk from A to C
Move 2th disk from A to B
Move 1th disk from C to B
Move 3th disk from A to C
Move 1th disk from B to A
Move 2th disk from B to C
Move 1th disk from A to C
```

B A
A B C
2
B
C
A
B
1
2
3
TO(0---)
S1 ꝛ
TO(---)
TOH(1,B,C,A)
Sysoc
TOH(1,A,B,C)
TOH(2,A,C,B)
Sysoc
TOH(2,B,A,C)
TOH(3,A,B,C)
main
N=1
src=A
hlp=B
dis=C
N=2
src=B
hlp=A
des=C
N=3
src=A
hlp=B
des=C